

Geometry Unit 12 Assessment Guide

General Information	The clarification for benchmark MA.912.GR.6.1 states, "Problems include relationships between two chords; two secants; a secant and a tangent; and the length of the tangent from a point to a circle." These relationships can be difficult for a student to remember. The lessons in the unit help a student to discover these relationships but a student may struggle separating them for a unit assessment. If your students appear to miss the same type of relationship, then consider looking at student work to determine if most of your students make the same error. This could help you discern how to support student learning. Clarification 1 of MA.912.GR.7.1 states, "Instruction includes using the Pythagorean Theorem and completing the square." Clarification 2 states, "Within the Geometry course, key features are limited to radius, diameter and the center." Key features and completing the square are concepts students learned in Algebra 1. If students struggle with either of these concepts, then consider supporting students by reviewing the key features of a line in unit 4 of Algebra 1. Students may also find reviewing key features of quadratic functions in unit 8 of Algebra 1 helpful. Students who struggle with completing the square may find that lessons 5 and 6 of unit 8 are beneficial. Clarification 3 of MA.912.GR.7.3 states, "Within the Geometry course, notations for domain and range are limited to inequality and set builder." For students who need support on these notations consider using Algebra 1 Unit 4 Lesson 1. Students may choose to solve the problems using a variety of methods. Consider following student work to consider whether minor errors demonstrate a student's knowledge of the relationships in a circle. If a student consistently rounds too early in finding a solution, rounds to the wrong place value, or repeats another common error then consider the totality of the error and make a single overall deduction rather than multiple deductions.
Calculator Information	The questions on the unit assessment are calculator.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1	Benchmark(s)	MA.912.GR.6.1
	☐ $\overline{ND} \cong \overline{WK}$ ☐ $\overline{NY} \cong \overline{XK}$ ☐ $\overline{BR} \cong \overline{BL}$ ☐ $\overline{DK} \cong \overline{KH}$ ☐ $\overline{NS} \cong \overline{SG}$ ☐ $\overline{GH} \cong \overline{ND}$	Recommended Scoring (6 Total Points)	Consider giving students 2 points for each correct answer. If students have any incorrect choices, then consider deducting 2 points for each incorrect answer.
Question	2	Benchmark(s)	MA.912.GR.7.3
		Recommended Scoring (6 Total Points)	Consider giving students 3 points for the correct center and 3 points for the correct radius length.

Question	3	Benchmark(s)	MA.912.GR.6.1
$k =$ 9.7	Recommended Scoring (7 Total Points)	Consider giving students 3 points if they correctly set up a correct equation. Consider giving 2 points for correct work in solving the equation and 2 points for the correct value of k .	
Question	4	Benchmark(s)	MA.912.GR.6.1
$x =$ 17.1	Recommended Scoring (7 Total Points)	Consider giving students 3 points if they correctly set up a correct equation. Consider giving 2 points for correct work in solving the equation and 2 points for the correct value of x .	

Question	5	Benchmark(s)	MA.912.GR.6.1
$b =$ 110		Recommended Scoring (7 Total Points)	Consider giving students 3 points if they correctly set up a correct equation. Consider giving 2 points for correct work in solving the equation and 2 points for the correct value of b .
Question	6	Benchmark(s)	MA.912.GR.6.1
Diameter 22.4		Recommended Scoring (7 Total Points)	Consider giving students 3 points if they correctly set up a correct equation or show their work in a different way. Consider giving 2 points for the correct radius and 2 points for the correct diameter.
Question	7	Benchmark(s)	MA.912.GR.7.2
Equation $(x + 3)^2 + (y - 3)^2 = 25$		Recommended Scoring (6 Total Points)	Consider distributing the points as 2 points for $(x + 3)^2$, 2 points for $(y - 3)^2$, and 2 points for 5^2 or 25.

Question		8a	Benchmark(s)	MA.912.GR.7.2						
<table><tr><th>Segment</th><th>Length</th></tr><tr><td>$\overline{HT}$</td><td>$y - 8$</td></tr><tr><td>$\overline{NT}$</td><td>$x - 5$</td></tr></table>			Segment	Length	$\overline{HT}$	$ y - 8 $	$\overline{NT}$	$ x - 5 $	Recommended Scoring (4 Total Points)	Students may write the length without the absolute value. Each teacher may emphasize this differently during instruction. Students may also write the expression for the length of each as $ 8 - y $ or $ 5 - x $.
Segment	Length									
$\overline{HT}$	$ y - 8 $									
$\overline{NT}$	$ x - 5 $									
Question		8b	Benchmark(s)	MA.912.GR.7.2						
<table><tr><th>Equation</th></tr><tr><td>$(x - 5)^2 + (y - 8)^2 = r^2$</td></tr></table>			Equation	$(x - 5)^2 + (y - 8)^2 = r^2$	Recommended Scoring (6 Total Points)	Consider giving students full credit if a student makes an error in 8a but correctly writes the equation of the circle based on that error. Consider using 2 points for each $(x - 5)^2$, $(y - 8)^2$ and r^2 .				
Equation										
$(x - 5)^2 + (y - 8)^2 = r^2$										

Question	8c	Benchmark(s)	MA.912.GR.7.2
The equation that shows the relationships of the side lengths of $\triangle HTN$ also represents the equation of circle N because point ☐ H ☐ T ☐ N is a representative point of ☐ $\triangle HTN$ ☐ circle N and can be placed anywhere on the ☐ circle. ☐ triangle. Any ☐ circle ☐ right triangle where the ☐ radius ☐ hypotenuse is the ☐ radius of the circle ☐ hypotenuse of the triangle will result in the same side lengths. <i>The last sentence could also correctly read, "Any <u>right triangle</u> where the <u>hypotenuse</u> is the <u>radius of the circle</u> will result in the same side lengths."</i>		Recommended Scoring (6 Total Points)	Consider assigning points as 3 points for the first sentence and 3 points for the second sentence. It is not suggested that points be further divided into 1 point for each choice as students may just be randomly making a choice. There are two correct ways to complete the last statement, depending on which choice students make in the first of three boxes in that statement. Both are shown in the key. This reflects what could happen on statewide assessments. Just a reminder that on statewide assessments students only earn one point for all parts being correct unlike the division of points suggested in the recommended scoring.

Question	9	Benchmark(s)	MA.912.GR.6.1
$x =$ 9		Recommended Scoring (7 Total Points)	Students may choose to do this problem in different ways. Consider giving students 3 points if they correctly set up a correct equation. Consider giving 2 points for correct work in solving the equation and 2 points for the correct value of x .
Question	10a	Benchmark(s)	MA.912.GR.6.1
$KH =$ 7.5		Recommended Scoring (7 Total Points)	Consider giving students 3 points if they apply a correct relationship. Consider giving 2 points for correct work in finding the value and 2 points for the correct length.
Question	10b	Benchmark(s)	MA.912.GR.6.1
$YB =$ 1.6		Recommended Scoring (7 Total Points)	Consider giving students 3 points if they apply a correct relationship. Consider giving 2 points for correct work in finding the value and 2 points for the correct length.

Question	11	Benchmark(s)	MA.912.GR.7.3																
<table border="1"> <tr> <th colspan="2">Center-Radius form</th><th>Center</th><th>Diameter</th></tr> <tr> <td colspan="2">work</td><td>$(4, -7)$</td><td>24</td></tr> <tr> <th>Equation</th><td>$(x - 4)^2 + (y + 7)^2 = 144$</td><th>Domain</th><th>Range</th></tr> <tr> <td colspan="2"></td><td>$-8 \leq x \leq 16$</td><td>$-19 \leq y \leq 5$</td></tr> </table>		Center-Radius form		Center	Diameter	work		$(4, -7)$	24	Equation	$(x - 4)^2 + (y + 7)^2 = 144$	Domain	Range			$-8 \leq x \leq 16$	$-19 \leq y \leq 5$	Recommended Scoring (17 Total Points)	Consider distributing the points as: • 6 points for correctly completing the square • 3 points for the correct equation • 2 points for the correct center • 2 points for the correct diameter • 2 points for the correct domain • 2 points for the correct range.
Center-Radius form		Center	Diameter																
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